Experiment no: 09

Aim: Implementation a Name Resolution protocol.

Theory:

Name Resolution Protocol in Distributed Systems

In a distributed system, a Name Resolution Protocol translates human-readable names or addresses into machine-readable network addresses.This enables processes or nodes within the system to communicate seamlessly using recognizablenames instead of manually handling IP addresses. Name resolution plays a crucial role in ensuring efficient resource access, service discovery, and network communication.

Common Name Resolution Protocols

1. Domain NameSystem(DNS)
    - DNS is the most widelyused name resolution protocol on the internet.
    - It maps domain names(e.g., example.com) to their corresponding IP addresses (e.g., 192.168.1.1).
    - DNS servers operate ina hierarchical manner, ensuring efficient resolution of queries.
    - Supports both forward resolution (name to IP) and reverse resolution (IP to name).
2. Simple Service DiscoveryProtocol (SSDP)
    - SSDP is used by network devices for service discovery and communication.
    - Commonly used in Universal Plug and Play (UPnP) environments.
    - Allows devices to advertise their services and dynamically discover other devices on the network.
    - Works using multicast messages to broadcast availability.
3. ServiceLocation Protocol (SLP)
    - SLP is designed to help clients locate network services efficiently.
    - Allows service discovery based on specific criteria like service type, location, or protocol.
    - Commonly used in enterprise networks and large distributed systems.
    - Operates without requiring prior knowledge of available services.
4. Lightweight Directory Access Protocol (LDAP)
    - LDAP is a protocol used to access and manage directory services.
    - Often utilized in enterprise environments to store information about users, groups, and network resources.
    - Enables centralized authentication and access control.

- Uses a hierarchical directory structure for efficient organization of information.

5. NetworkInformation Service (NIS)
   - NIS is a protocol used in Unix-based systems for name resolution and authentication.
   - It allows clients to retrieve network-related data such as user accounts, hostnames, and passwordsfrom a central server.
   - Primarily used in enterprise environments for managing large-scale networks.
   - Provides a simplified way to share configuration and authentication data across multiple systems.

labdate.py

```python
import socket

def get_ip_address(url):
    try:
        host_ip = socket.gethostbyname(url)
        print("Hostname:", url)
        print("IP:", host_ip)
    except socket.gaierror:
        print("Unable to get hostname and IP")

if __name__ == "__main__":
    url = "labdate2023-24.netlify.app"
    get_ip_address(url)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\LENOVO\OneDrive\Desktop\sma exp 10> python labdate.py
Hostname: labdate2023-24.netlify.app
IP: 52.76.120.174
PS C:\Users\LENOVO\OneDrive\Desktop\sma exp 10>
```

Conclusion

This experiment demonstrated how to resolve domain names into IP addresses using Python's socket module. It provided insights into the working of DNS and the importance of name resolution in networking.

```
import socket

def get_ip_address(url):
    try:
        # gethostbyname() performs the DNS lookup
        host_ip = socket.gethostbyname(url)

        print("Hostname:", url)
        print("IP Address:", host_ip)

    except socket.gaierror:
        # This error triggers if the URL is invalid or the site is
down
        print("Error: Unable to get hostname and IP")

if __name__ == "__main__":
    # The URL provided in your lab manual
    url = "labdate2023-24.netlify.app"
    ge
t_ip_address(url)
```